Sohini Dutta

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Research Interests: Visual representation learning, 3D scene understanding, video understanding, spatial and physical reasoning, multimodal learning, and object-centric representations.

Education

Indian Institute of Technology Madras
2019 - 2024

Dual Degree - B.Tech. in Mechanical Engineering + M.Tech. in Computational Engineering **CGPA: 8.88/10**

Research Experience

Project Associate, Indian Institute of Science
Sep '25 - Present

Advisor: **Dr. Prathosh A. P.**, Department of Electrical Communication Engineering

ArtiCollision: Predicting the Consequences of Articulated Motion (first author, manuscript)

- Initiated and currently lead a study on whether MLLMs predict if opening a door or drawer causes a collision.
- Built a SAPIEN + PartNet-Mobility generator of matched single-image scenes where only the obstacle moves.
- Fine-tuning raises accuracy, yet evaluated MLLMs struggle to separate collisions from 5–20 mm near misses.
- Controlled tests show sensitivity to obstacle-position cues and limited gains from structure supervision.
- Error analysis of a diagnostic image-to-geometry pipeline identifies segmentation and depth estimation as its main error sources, while the motion estimate contributes little.

One Label per Region: Superpixel Tokenization for Decoder-Free MLLM Segmentation (second author)

Linked manuscript describes the earlier version; an updated manuscript is in preparation.

- Co-developed SPARSE (linked manuscript): a decoder-free MLLM that labels superpixel tokens as text.
- Co-designed, built, and ablated the mask-based positional encoding; contributed geometry-aware pooling.
- Built region-level supervision and the LLaVA-1.5 training and ADE20K evaluation pipeline.
- Cuts ∼1024 per-patch output labels to ∼64–128 region labels; 19.1 vs. 16.5 mIoU (Text4Seg) on ADE20K.
- Ongoing extension beyond the linked manuscript: a learned readout of the MLLM's final-layer hidden states selects superpixels, improving referring-expression segmentation.

Undergraduate Researcher, IIT Madras
Jul '22 - Nov '22

Advisor: **Dr. P. V. Manivannan**

Monocular Path Perception for Autonomous Navigation, AmigoBot

- Built a single-image RGB traversability pipeline: MiDaS depth, SLIC superpixels, region-graph merging.
- Implemented in Python, OpenCV, and PyTorch as low-cost perception for robots without a depth sensor.

Other Major Projects

Directional Freeze Desalination (M.Tech. Thesis, IIT Madras) under **Dr. Sateesh Gedupudi** Aug '23 - Jun '24

- Modelled freeze desalination in ANSYS Fluent, coupling phase change, solute transport, and convection.
- Implemented and validated a Scheil solidification model; characterised freezing rate vs. salt rejection.

Cross-Laminated Timber Joints (UBC, Vancouver) under **Dr. Frank Lam** May '23 - Aug '23

- Ran monotonic and cyclic tests on timber joints; analysed load–displacement and fatigue in Python.

Operational Solutions of Heat Equations (IISc, Bengaluru) under **Dr. N. K. Ravichandran** May '22 - Jul '22

- Solved and compared Fourier, hyperbolic, and Guyer–Krumhansl heat-transport models in MATLAB.

Industry Experience

Software Developer, Uni Cards (Uniorbit Technologies)

Jun '24 - Jun '25

- Built distributed Java/Spring Boot services with Temporal workflows and MongoDB.
- Migrated customer onboarding state machines to durable workflows, reducing microservice failures.
- Built a mobile aggregation backend and a voucher-store backend; voucher purchases doubled after launch.

Technical Skills

ML / AI	PyTorch, Hugging Face, DeepSpeed, PEFT/LoRA, W&B, LLaVA, Qwen-VL, InternVL, CLIP
Vision	Segmentation, visual grounding, monocular depth estimation, SAM, superpixels, OpenCV
Simulation	SAPIEN, PartNet-Mobility, URDF, articulated kinematics, collision and contact analysis
Programming	Python, C/C++, Java, Spring Boot, MongoDB, NumPy, SciPy, MATLAB, ANSYS Fluent, Git, Linux

Key Coursework

Mathematics	Series & Matrices, Probability, Statistics & Stochastic Processes, Functions of Several Variables, Differential Equations, Numerical Techniques for Engineers
Computing	Computational Tools: Algorithms, Data Structures & Programs, High Performance Computing Lab, Finite Element Analysis, Introduction to Programming
Robotics	Kinematics & Dynamics of Machinery, Mechatronic Systems

Achievements

2024 Sri G. S. Bhuvaneshwar Prize: highest CGPA over two semesters, Interdisciplinary Dual Degree, IIT Madras.
2023 Selected for a research internship at the University of British Columbia through Mitacs Globalink.
2022 Awarded the IASc–INSA–NASI Summer Research Fellowship, Indian Institute of Science.
2018 Cleared the Regional Mathematics Olympiad (Homi Bhabha Centre for Science Education).
2017 Received the HRD Ministry's annual scholarship for clearing the National Talent Search Examination.

Teaching & Leadership

2024 Teaching Assistant at IIT Madras: Thermal Engineering Laboratory (HTTA scholarship).
2023 Teaching Assistant at IIT Madras: Mechanical Design Laboratory.
2021 Coordinator at Shaastra '21, IIT Madras: Shows & Exhibitions.
2020 Co-founder, TechTraction podcast: interviews including Dr. Hartmut Neven (Google Quantum AI).